

RARITONE BACKEND PROJECT REPORT

Internship Project Report

Submitted By:

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Internship Organization:

Raritone Private Limited

Duration:

1 Month

DECLARATION

I hereby declare that the project report entitled “Raritone Backend Project” submitted in partial fulfillment of the internship requirements is my original work. The information and results presented in this report are based on the work carried out during the internship period.

Signature: Meka Hrishi Teja Chowdary

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the management and technical team of Raritone for providing me with the opportunity to work on this project. Their guidance, support, and valuable insights helped me gain practical knowledge in backend development, API design, database management, and system integration.

I also extend my thanks to my mentors and team members for their continuous support throughout the internship period, which helped me understand real-world software development practices.

ABSTRACT

The Raritone Backend Project is a scalable and modular backend system designed for a virtual fashion platform that enhances the online shopping experience through product management, order processing, and virtual try-on capabilities. Traditional e-commerce systems often lack proper visualization features, leading to poor size selection and high return rates.

The main goal of this project was to build a secure, efficient, and scalable backend system using Node.js, Express.js, and MongoDB, following a modular architecture approach. This ensures better maintainability, flexibility, and future expansion.

The system includes core modules such as product management, order processing, and virtual try-on workflow support. It also integrates secure authentication, structured APIs, and optimized database design.

The backend is designed to support future AI-based enhancements such as intelligent recommendations, virtual try-on improvements, and personalized fashion experiences. Overall, the project provides a strong foundation for a modern, AI-ready fashion technology platform.

1. Introduction

The Raritone Backend System was developed as a scalable and modular backend solution for a virtual fashion platform. The system supports core functionalities such as product management, order processing, and virtual try-on workflows, along with preparation for future AI-driven features.

The backend was built using Node.js, Express.js, and MongoDB, following a modular architecture to ensure maintainability, scalability, and future expansion.

This internship provided practical exposure to backend development, REST API creation, database handling, and collaborative software development practices.

2. Project Objectives

The main objectives of the project were:

- Develop a modular backend architecture
- Build RESTful APIs for all core modules
- Implement product management functionality
- Support order processing workflows
- Design virtual try-on system structure
- Ensure scalable database design
- Prepare system for future AI integration

3. My Contributions

During the internship, I contributed to the development, design, and testing of multiple backend modules.

Major Contributions:

- Product Module Development
- Product API Development
- Product Management System
- Order Module Development
- Virtual Try-On Module Development
- Try-On Workflow Design
- Backend Testing Support
- GitHub Collaboration

4. Product Module Development

I contributed to building the Product Module, which handles all product-related operations in the system.

Features Implemented:

- Create Product
- Retrieve Products
- Update Product Details
- Delete Product
- Manage Product Information

APIs Developed:

- POST /api/products
- GET /api/products
- GET /api/products/:id
- PUT /api/products/:id
- DELETE /api/products/:id

Key Learnings:

- REST API design
- CRUD operations
- MongoDB schema handling
- Backend validation techniques

5. Product Management System

The Product Management System acts as the core catalog system of the platform.

Responsibilities:

- Maintaining product data
- Managing inventory-related information
- Supporting future recommendation systems
- Organizing product structure

Benefits:

- Centralized product database
- Easy management of product data
- Scalable for future features

6. Order Module Contribution

I contributed to the Order Module, which handles customer purchase workflows.

Features:

- Order creation
- Order retrieval
- Order updates
- Order tracking

Importance:

The module ensures proper management of customer transactions and purchase history.

Key Learnings:

- Data relationships between users and orders
- Backend workflow handling
- Order lifecycle management

7. Virtual Try-On Module Development

I worked on designing and contributing to the Virtual Try-On Module.

Purpose:

To enable future AI-powered fashion visualization where users can try products virtually before purchasing.

Planned Features:

- Product selection system
- User image upload handling
- Avatar-based try-on support
- AI integration readiness
- Result storage system

Key Learnings:

- AI workflow understanding
- Image processing concepts
- Future system design planning

8. Try-On Workflow Design

A structured workflow was designed to support future AI integration.

Workflow:

User selects product

Uploads image / avatar

Try-on request generated

AI processing layer

Generated preview result

Displayed to user

Benefits:

- Better user experience
- Improved purchase confidence
- Reduced product return rates

9. Database Interaction Experience

Worked extensively with MongoDB and Mongoose during development.

Concepts Learned:

- Collections and documents
- Schema design
- ObjectId relationships
- Data retrieval techniques
- Database relationships

Outcome:

Improved understanding of real-world database-driven backend systems.

10. API Development Experience

Developing APIs helped me understand backend communication systems.

Skills Developed:

- Route creation
- Controller implementation
- Request handling
- Response formatting
- Error handling

Outcome:

Gained confidence in building production-level REST APIs.

11. GitHub Collaboration

The project was developed collaboratively using GitHub.

Activities:

- Branch management
- Commit handling
- Code integration
- Version control
- Team synchronization

Learning Outcomes:

- Team-based development workflow
- Version control best practices
- Collaboration efficiency

12. Challenges Faced

Understanding Virtual Try-On Workflow

Since virtual try-on systems involve AI concepts, understanding the complete workflow required additional research.

Solution: Studied existing systems and implementation approaches.

Managing Product Relationships

Handling structured product data for future recommendation systems required careful design.

Solution: Used scalable schema design and MongoDB relationships.

Module Integration

Ensuring smooth interaction between multiple backend modules was challenging.

Solution: Performed testing and validation during integration.

13. Key Learnings

- Backend development fundamentals
- REST API design and implementation
- MongoDB database management
- Product and order system workflows
- Virtual try-on architecture planning
- GitHub collaboration
- Team-based development practices

14. Future Improvements

- AI-powered virtual try-on system
- Recommendation engine integration
- Advanced product mapping system
- Personalized fashion suggestions
- Real-time updates using WebSockets
- Cloud-based scaling improvements

15. Conclusion

The Raritone Summer Internship provided valuable hands-on experience in backend development and real-world software engineering practices.

Through contributions to the Product Module, Order Module, and Virtual Try-On Module, I gained strong practical knowledge in API development, database design, workflow structuring, and collaborative development.

This internship helped strengthen my technical skills, problem-solving ability, and understanding of scalable backend systems, preparing me for future opportunities in software development.